



MAT-121

Chapter: **Partial Differentiation**

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Partial Differentiation

Introduction: The **partial derivative** is used in vector calculus and differential geometry.

In Mathematics, sometimes the function depends on two or more variables. Here, the derivative converts into the partial derivative since the function depends on several variables. In this article, We will learn about the definition of partial derivatives, their formulas, partial derivative rules.

What is Partial Differentiation?

Suppose, we have a function $u = f(x, y)$, which depends on two variables x and y , where x and y are independent of each other. Then the result of differentiating u with respect to x , treating y as a constant, is called the partial derivative of u , with respect to x , and is denoted by the symbols $\frac{\partial u}{\partial x}, \frac{\partial f}{\partial x}, u_x$ or f_x .

$$\text{Analytically, } f_x = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h};$$

And partial derivative of function f with respect to y , keeping x as a constant, we get

$$f_y = \frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} \text{ When this limit exist.}$$

Problem-1:

If $u = x^2 + xy + y^2$, then find u_x & u_y .

Solution: Given, $u = x^2 + xy + y^2$ (1)

Differentiating Equ.(1) partially w.r.t. x and y respectively, we get

$$\therefore u_x = \frac{\partial u}{\partial x} = 2x + 1.y + 0$$

$$\therefore u_x = 2x + y$$

$$\therefore u_y = \frac{\partial u}{\partial y} = 0 + x.1 + 2y$$

$$\therefore u_y = x + 2y$$

Ans: $u_x = 2x + y$ and $u_y = x + 2y$

Problem-2: If $u = yz + zx + xy$, then find u_x, u_y & u_z .

Solution: Given $u = yz + zx + xy$ (1)

Differentiating Equ.(1) partially w.r.t. x, y and z respectively, we get

$$\begin{aligned}\therefore u_x &= \frac{\partial u}{\partial x} = 0 + z \cdot 1 + 1 \cdot y + 0 \\ &\Rightarrow u_x = y + z\end{aligned}$$

$$\begin{aligned}\therefore u_y &= \frac{\partial u}{\partial y} = z \cdot 1 + 0 + x \cdot 1 \\ &\Rightarrow u_y = x + z\end{aligned}$$

$$\begin{aligned}\therefore u_z &= \frac{\partial u}{\partial z} = y \cdot 1 + x \cdot 1 + 0 \\ &\Rightarrow u_z = x + y.\end{aligned}$$

Second order partial derivatives:

$$\triangleright u_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial x^2} ,$$

$$\triangleright u_{yy} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial^2 u}{\partial y^2} ,$$

$$\triangleright u_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial x \partial y}$$

$$\triangleright f_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} ,$$

$$\triangleright f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x \partial y}$$

Problem-3:

Calculate the values of $f_x, f_y, f_{xx}, f_{yy}, f_{xy}$ of the following functions:

i) $f(x, y) = x^3 + x^2y^2 + y^3.$

ii) $f(x, y) = x^3 + 3x^2y + 3xy^2 + y^3.$

iii) $f(x, y) = x\cos y + y\cos x.$

iv) $f(x, y) = ax^2 + 2hxy + by^2.$

Solution: i) Given $f(x, y) = x^3 + x^2y^2 + y^3 \dots \dots \dots (1)$

Differentiating equation (1) partially with respect to x & y respectively,

$$\therefore f_x = \frac{\partial f}{\partial x} = 3x^2 + 2xy^2 + 0 = 3x^2 + 2xy^2$$

$$\therefore f_y = \frac{\partial f}{\partial y} = 0 + x^2 \cdot 2y + 3y^2 = 2x^2y + 3y^2$$

$$\therefore f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = 3 \cdot 2x + 2 \cdot 1 \cdot y^2 = 6x + 2y^2$$

Problem-3: Cont...

$$\therefore f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = 2x^2 \cdot 1 + 3 \cdot 2 \cdot y = 2x^2 + 6y$$

$$\therefore f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = 0 + 2x \cdot 2y = 4xy$$

ii) Given $f(x, y) = x^3 + 3x^2y + 3xy^2 + y^3 \dots \dots (1)$

Differentiating equation (1) partially with respect to x & y respectively,

$$\therefore f_x = \frac{\partial f}{\partial x} = 3x^2 + 3 \cdot 2xy + 3 \cdot 1 \cdot y^2 + 0 = 3x^2 + 6xy + 3y^2$$

$$\therefore f_y = \frac{\partial f}{\partial y} = 0 + 3x^2 \cdot 1 + 3x \cdot 2y + 3y^2 = 3x^2 + 6xy + 3y^2$$

$$\therefore f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = 3 \cdot 2x + 6 \cdot 1 \cdot y + 0 = 6x + 6y$$

$$\therefore f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = 0 + 6x \cdot 1 + 3 \cdot 2y = 6x + 6y$$

$$\therefore f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = 0 + 6x \cdot 1 + 3 \cdot 2y = 6x + 6y.$$

Problem-3: Cont...

iii) Given $f(x, y) = x \cos y + y \cos x \dots \dots \dots (1)$

Differentiating equation(1) partially with respect to x & y respectively,

$$\therefore f_x = \frac{\partial f}{\partial x} = 1 \cdot \cos y + y(-\sin x) = \cos y - y \sin x$$

$$\therefore f_y = \frac{\partial f}{\partial y} = x(-\sin y) + 1 \cdot \cos x = -x \sin y + \cos x$$

$$\therefore f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = 0 - y \cos x = -y \cos x$$

$$\therefore f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = -x \cos y + 0 = -x \cos y$$

$$\begin{aligned} \therefore f_{xy} &= \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = -\sin y - 1 \cdot \sin x \\ &= -(\sin x + \sin y). \text{ (Answer)} \end{aligned}$$

Problem-4:

- i) If $u = x^2 + y^2 + z^2$, then show that $xu_x + yu_y + zu_z = 2u$.
- ii) If $u = x^2y + y^2z + z^2x$, then show that

$$u_x + u_y + u_z = (x + y + z)^2.$$

- iii) Evaluate $xV_x + yV_y + zV_z$, where $V = \frac{x}{y} + \frac{z}{x} + \frac{y}{z}$.

Solution: i) Given $u = x^2 + y^2 + z^2$ (1)

Differentiating equation (1) partially with respect to x, y & z respectively,

$$\therefore u_x = \frac{\partial u}{\partial x} = 2x + 0 + 0 = 2x$$

$$\therefore u_y = \frac{\partial u}{\partial y} = 0 + 2y + 0 = 2y$$

$$\text{and } u_z = \frac{\partial u}{\partial z} = 0 + 0 + 2z = 2z$$

Problem-4: Cont...

$$\begin{aligned}\text{Thus } xu_x + yu_y + zu_z &= x.2x + y.2y + z.2z \\ &= 2x^2 + 2y^2 + 2z^2 \\ &= 2(x^2 + y^2 + z^2) = 2u\end{aligned}$$

$$\text{So, } xu_x + yu_y + zu_z = 2u.$$

Hence proved.

$$\text{ii) Given } u = x^2y + y^2z + z^2x \quad (1)$$

Differentiating equation (1) partially with respect to x, y & z respectively,

$$\therefore u_x = \frac{\partial u}{\partial x} = 2xy + 0 + z^2.1 = 2xy + z^2$$

$$\therefore u_y = \frac{\partial u}{\partial y} = x^2.1 + 2yz + 0 = x^2 + 2yz$$

Problem-4: Cont...

$$\text{and } u_z = \frac{\partial u}{\partial z} = 0 + y^2 \cdot 1 + 2zx = y^2 + 2zx$$

$$\begin{aligned}\text{So, } u_x + u_y + u_z &= 2xy + z^2 + x^2 + 2yz + y^2 + 2zx \\ &= x^2 + y^2 + z^2 + 2xy + 2yz + 2zx \\ &= (x + y + z)^2\end{aligned}$$

$$\text{Thus, } u_x + u_y + u_z = (x + y + z)^2.$$

Hence proved.

Problem-4: Cont...

iii) Given $V = \frac{x}{y} + \frac{z}{x} + \frac{y}{z}$ (1)

Differentiating equation (1) partially with respect to x , y & z respectively,

$$\therefore V_x = \frac{\partial V}{\partial x} = \frac{1}{y} + z \cdot \left(-\frac{1}{x^2} \right) + 0 = \frac{1}{y} - \frac{z}{x^2}$$

$$\therefore V_y = \frac{\partial V}{\partial y} = x \cdot \left(-\frac{1}{y^2} \right) + 0 + \frac{1}{z} = -\frac{x}{y^2} + \frac{1}{z}$$

$$\text{and } V_z = \frac{\partial V}{\partial z} = 0 + \frac{1}{x} + y \left(-\frac{1}{z^2} \right) = \frac{1}{x} - \frac{y}{z^2}$$

$$\begin{aligned} \text{So, } xV_x + yV_y + zV_z &= x \left(\frac{1}{y} - \frac{z}{x^2} \right) + y \left(-\frac{x}{y^2} + \frac{1}{z} \right) + z \left(\frac{1}{x} - \frac{y}{z^2} \right) \\ &= \frac{x}{y} - \frac{z}{x} - \frac{x}{y} + \frac{y}{z} + \frac{z}{x} - \frac{y}{z} = 0 \end{aligned}$$

$$\therefore xV_x + yV_y + zV_z = 0.$$

(Answer)

Since $\frac{d \frac{1}{x^n}}{dx} = \frac{dx^{-n}}{dx} = -nx^{-n-1}$.

$$\therefore \frac{d}{dx} \left(\frac{1}{x} \right) = \frac{d(x^{-1})}{dx} = -x^{-1-1}$$
$$= -\frac{1}{x^2}$$

Problem-5:

Evaluate $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$; where $u = \sin^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x}$.

Solution: Given $u = \sin^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x}$ (1)

Differentiating equation (1) partially with respect to x & y respectively,

$$\begin{aligned} \therefore \frac{\partial u}{\partial x} &= \frac{1}{\sqrt{1-\frac{x^2}{y^2}}} \cdot \left(\frac{1}{y}\right) + \frac{1}{1+\frac{y^2}{x^2}} \cdot y \cdot \left(-\frac{1}{x^2}\right) = \frac{1}{\sqrt{\frac{y^2-x^2}{y^2}}} \cdot \frac{1}{y} - \frac{1}{\frac{x^2+y^2}{x^2}} \cdot \frac{y}{x^2} \\ &= \frac{y}{\sqrt{y^2-x^2}} \cdot \frac{1}{y} - \frac{1}{\frac{x^2+y^2}{x^2}} \cdot \frac{y}{x^2} \end{aligned}$$

or,
$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{y^2-x^2}} - \frac{y}{x^2+y^2}$$

Problem-5: Conti....

$$\begin{aligned}\therefore \frac{\partial u}{\partial y} &= \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \cdot x \cdot \left(-\frac{1}{y^2} \right) + \frac{1}{1 + \frac{y^2}{x^2}} \cdot \left(\frac{1}{x} \right) \\ &= -\frac{1}{\sqrt{\frac{y^2 - x^2}{y^2}}} \cdot \frac{x}{y^2} + \frac{1}{\frac{x^2 + y^2}{x^2}} \cdot \frac{1}{x} = \frac{-y}{\sqrt{y^2 - x^2}} \cdot \frac{x}{y^2} + \frac{1}{\frac{x^2 + y^2}{x^2}} \cdot \frac{1}{x}\end{aligned}$$

$$\text{or, } \frac{\partial u}{\partial y} = \frac{-x}{y\sqrt{y^2 - x^2}} + \frac{x}{x^2 + y^2}$$

$$\begin{aligned}\text{So, } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= x \left(\frac{1}{\sqrt{y^2 - x^2}} - \frac{y}{x^2 + y^2} \right) + y \left(\frac{-x}{y\sqrt{y^2 - x^2}} + \frac{x}{x^2 + y^2} \right) \\ &= \frac{x}{\sqrt{y^2 - x^2}} - \frac{xy}{x^2 + y^2} - \frac{x}{\sqrt{y^2 - x^2}} + \frac{xy}{x^2 + y^2} = 0\end{aligned}$$

$$\text{Thus } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0. \text{ (Answer)}$$

Problem-6:

If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then show that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}.$$

Solution: Given $u = \log(x^3 + y^3 + z^3 - 3xyz)$ (1)

Differentiating equation (1) partially with respect to x , y & z respectively,

$$\begin{aligned}\therefore \frac{\partial u}{\partial x} &= \frac{1}{x^3 + y^3 + z^3 - 3xyz} (3x^2 + 0 + 0 - 3 \cdot 1 \cdot yz) \\ &= \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz} \\ \therefore \frac{\partial u}{\partial y} &= \frac{1}{x^3 + y^3 + z^3 - 3xyz} (0 + 3y^2 + 0 - 3x \cdot 1 \cdot z) \\ &= \frac{3y^2 - 3xz}{x^3 + y^3 + z^3 - 3xyz}\end{aligned}$$

Problem-6 Conti...

$$\text{and } \frac{\partial u}{\partial z} = \frac{1}{x^3 + y^3 + z^3 - 3xyz} (0 + 0 + 3z^2 - 3xy \cdot 1)$$

$$\text{or, } \frac{\partial u}{\partial z} = \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz}$$

$$\begin{aligned} \text{So, } \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz} + \frac{3y^2 - 3xz}{x^3 + y^3 + z^3 - 3xyz} + \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz} \\ &= \frac{3x^2 - 3yz + 3y^2 - 3xz + 3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz} \\ &= \frac{3(x^2 + y^2 + z^2 - yz - xz - xy)}{(x+y+z)(x^2 + y^2 + z^2 - yz - xz - xy)} = \frac{3}{x+y+z} \end{aligned}$$

$$\text{Thus } \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}.$$

Hence Showed.

Problem-7:

If $u = \log(x^2 + y^2)$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Solution: Given $u = \log(x^2 + y^2) \dots \dots \dots (1)$

Differentiating equation (1) partially with respect to x & y respectively,

$$\therefore \frac{\partial u}{\partial x} = \frac{1}{x^2 + y^2} (2x + 0) = \frac{2x}{x^2 + y^2}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{2x(2x + 0) - (x^2 + y^2) \cdot 2}{(x^2 + y^2)^2}$$

$$= \frac{4x^2 - 2x^2 - 2y^2}{(x^2 + y^2)^2}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$$

$$\therefore \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{u \frac{dv}{dx} - v \frac{du}{dx}}{v^2}.$$

Problem-5: Conti....

$$\text{And } \frac{\partial u}{\partial y} = \frac{1}{x^2 + y^2} (0 + 2y) = \frac{2y}{x^2 + y^2}$$

$$\begin{aligned} \therefore \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{2y(0 + 2y) - (x^2 + y^2) \cdot 2}{(x^2 + y^2)^2} \\ &= \frac{4y^2 - 2x^2 - 2y^2}{(x^2 + y^2)^2} \end{aligned}$$

$$\therefore \frac{\partial^2 u}{\partial y^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}$$

$$\begin{aligned} \text{So, } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= \frac{2x^2 - 2y^2}{(x^2 + y^2)^2} + \frac{2y^2 - 2x^2}{(x^2 + y^2)^2} \\ &= \frac{2x^2 - 2y^2 + 2y^2 - 2x^2}{(x^2 + y^2)^2} = 0 \end{aligned}$$

$$\text{Thus } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \text{ Hence Shown.}$$

Problem-8:

If $u = (x^2 + y^2 + z^2)^{\frac{1}{2}}$, then show that $u_{xx} + u_{yy} + u_{zz} = 2/u$.

Problem-9:

i) If $u = (ax + by)^2 - (x^2 + y^2)$ and $a^2 + b^2 = 2$, then show that $u_{xx} + u_{yy} = 0$.

ii) If $u = 4(\alpha x + \beta y + \gamma z)^2 - (x^2 + y^2 + z^2)$ and $\alpha^2 + \beta^2 + \gamma^2 = \pi$, then find the value of $u_{xx} + u_{yy} + u_{zz}$.

THANKS TO ALL